bricks of all matter. An atom consists of a central nucleus of positive charge around which small negatively charged particles, called electrons revolved in different paths or orbits.

Nucleus is the central part of an atom which contain protons and neutrons. A proton is a positively charged particle, while the neutron has the same mass as the proton, but has no charge. Therefore, the nucleus of an atom is positively charged. So all the materials are in stable condition. The sum of protons and neutrons constitutes the entire weight of an atom and is called atomic weight. The number of electrons is equal to the number of protons in an atom and is called as atomic number.

i.e, Atomic number=number of protons (or)number of electrons.
Atomic weight=number of protons + number of neutrons.

10.1.2 Structure of Electron in An Atom

The electrons in an atom revolve around the nucleus in different orbits or paths. The number of electrons in any orbit is determined by the following rules.

The number of electrons in any orbit is given by $2n^2$ where n is the number of the orbit.

For example

i) First orbit contains = $2 \times 1^2 = 2$ electrons
ii) Second orbit contains = $2 \times 2^2 = 8$ electrons
iii) Third orbit contains = $2 \times 3^2 = 18$ electrons
iv) The last orbit cannot have more than 8 electrons
v) The last but one orbit cannot have more than 18 electron

10.1.3 Structure of Elements

We have seen that all atoms are made up of protons, neutrons and electrons. The difference between various types of element is due to the different number and arrangement of these particles within their atom. For example, the structure of copper atom is different from that of carbon atom and hence the two elements have different properties. (Ref Fig. 10.1)

The atomic structure can be easily formed if we know the atomic weight and atomic number of the element.

Ex:copper→Atomic weight = 64
Atomic number = 29,
Number of protons=Number of electrons=29,
And Number of neutrons=64–29=35

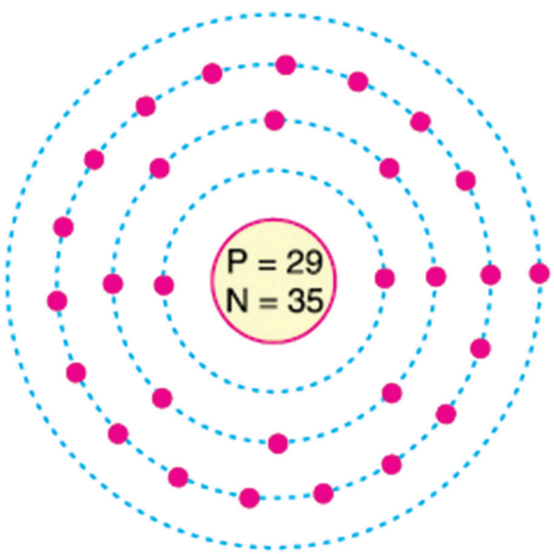


Fig 10.1 *Atomic Structure of Copper*

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